

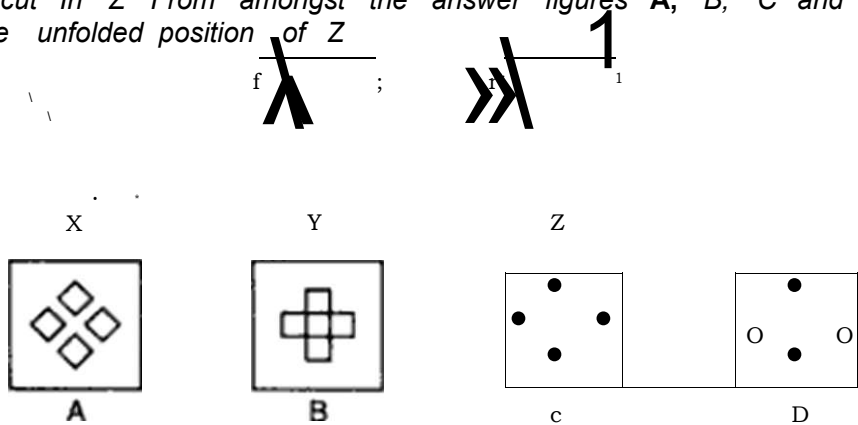
11. PAPER CUTTING

In this chapter we shall study the problems relating to the analysis of pattern that is formed when a folded piece of paper has been cut in a definite design.

Problems on Paper Cutting : In this type of questions, a set of three figures showing the manner in which a piece of paper has been folded, are being given. In each of the first two figures, a dotted line together with an arrow on it has been given. The dotted line is the reference line along which the paper has to be folded and the arrow indicates the direction of the fold. In the third figure, there are marks showing the position and the nature of the cut made in the folded sheet. The examinee has to select one of the figures from the set of four answer figures A, B, C and D which would most closely resemble the pattern when the paper is unfolded.

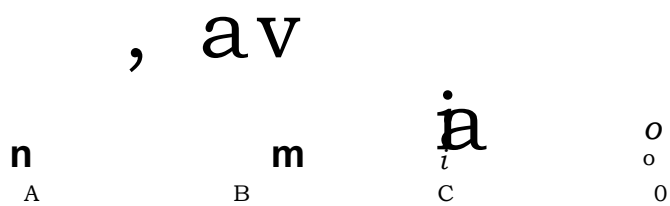
Remark : Evidently, the designs of the cut will appear on each one of the folds made in the paper.

Ex. 1. Consider the following three figures, marked X, Y, Z showing one fold in X another in Y and cut in Z. From amongst the answer figures A, B, C and D, select the one, showing the unfolded position of Z.



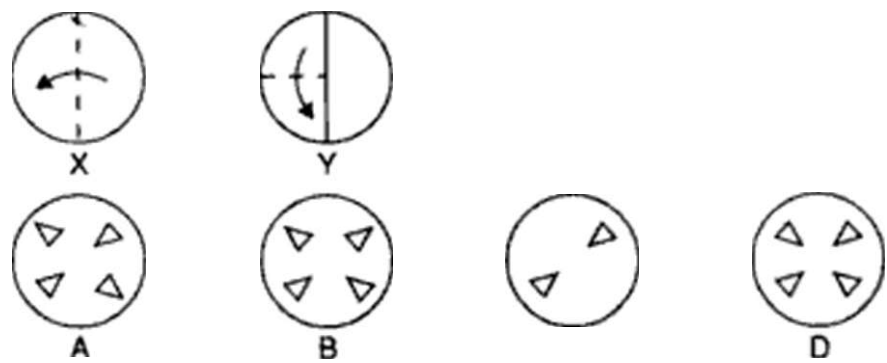
Sol. In fig. X, the upper triangular half of the paper has been folded over the lower half. In fig. Y, the paper is refolded to a quarter triangle. In fig. Z, a square has been punched in the folded paper. Clearly, the square will appear in each of the triangular quarters of the paper. Thus, when the paper is unfolded, four squares will appear symmetrically over it and it will resemble fig. (C).

Ex. 2. Consider the figures X and Y showing a rectangular sheet of paper folded in fig. X and punched in fig. Y. From amongst the answer figures A, B, C and D, select the figure, which will most closely resemble the unfolded position of fig. Y.



Sol. In fig. X, the right half of the rectangular paper sheet is folded over the left half. In fig. Y, two semicircles are punched into the folded paper. When the paper is unfolded, the semicircles in the two halves will join to form circles. Thus, two circles will appear in the unfolded position of fig. Y. Hence, fig. (D) is the correct answer.

Ex. 3. In the following question, three figures X, Y, Z, showing a sequence of folding a paper are given. The third figure depicts the cuts made in the folded paper. Select the figure from the answer figures marked A, B, C and D which would most closely resemble the third paper when unfolded.



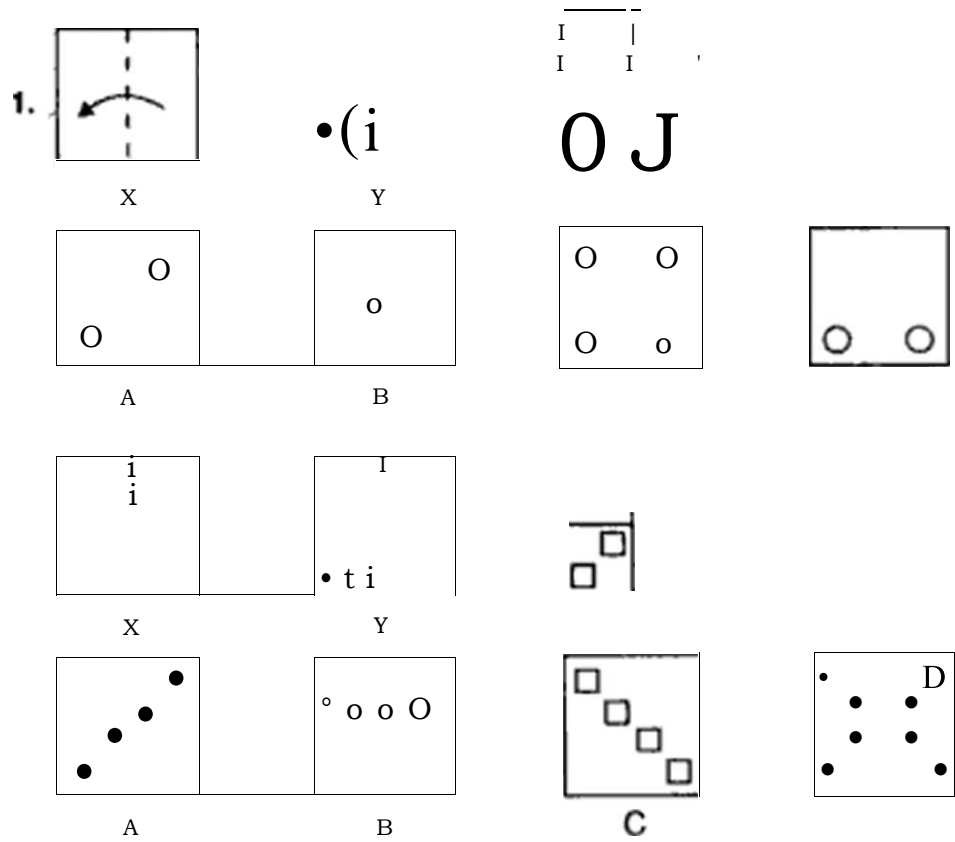
Sol. Here, the circular sheet of paper is one© folded along a diameter such that one semicircle lies above another. Now, the sheet is refolded along the line of symmetry such that all the quarter circles lie one above another. Then a triangular cut is made on the folded sheet. When this sheet is unfolded once, it will appear as shown below :

O

This sheet when completely unfolded will contain triangles on each quarter and wttl appear as fig. (0).
Hence, the answer is fig. (£7).

EXERCISE 11

Directions : The questions that follow contain a set of three figures X, Y and Z showing a sequence of folding of a piece of paper. Fig. (Z) shows the manner in which the folded paper has been cut These three figures are followed by four answer figures from which you have to choose a figure which would most closely resemble the unfolded form of fig. (Z).

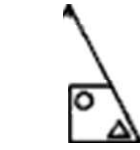


Hidden page

8.


X


B





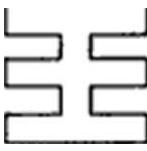
9.

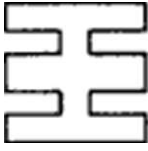
t

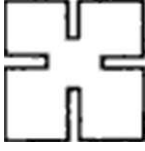
j g

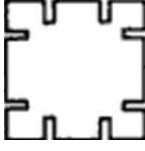
(Assistant Grade, 1993)

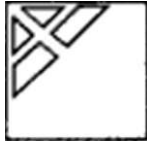
10.


B

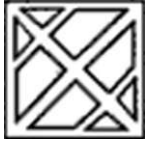


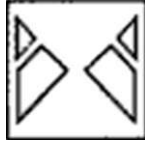





B







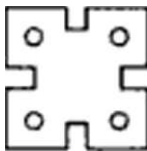
11.

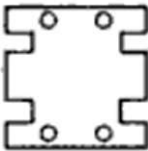

Y

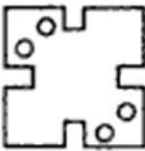

B

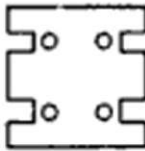

Z


D


B


Y


Z


D

12.


B


Y


Z


D

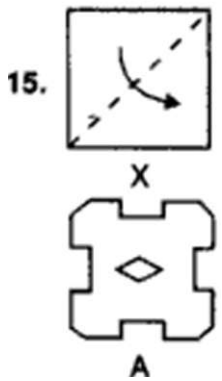
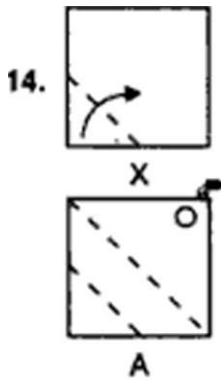
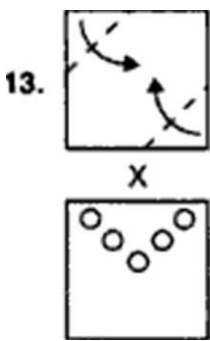

B


Y


Z

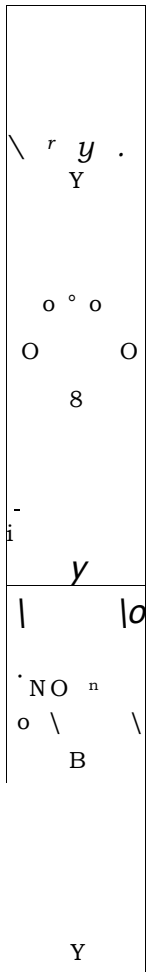
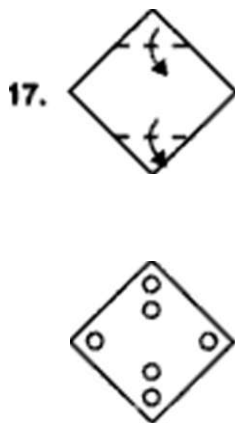

D

Copyrighted material



- C 3

C 2)



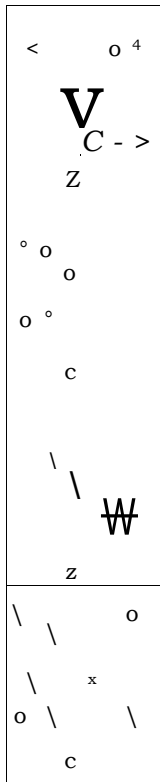
E

B

fr Y 1 /

Vftt

B



B

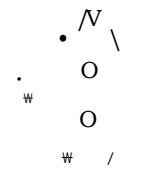
c

/ O N

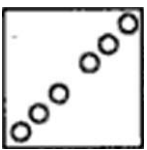
' o

V o /

7



(Assistant Grade, 1994)



(C.B.I. 1996)

8

D

18.

A

X

Y

Z

19.

X

A

B

A

B

(Auditor's Exam, 1991)

20.

X

A

B

21.

p

l v

/J A

(U.D.C. 1991)

22.

A

B

C

X

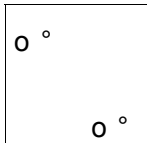
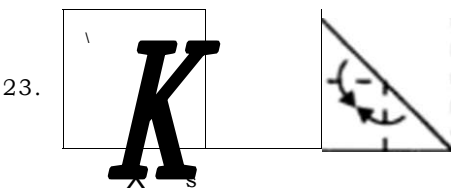
Y

Z

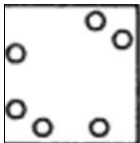
B

C

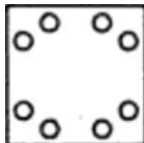
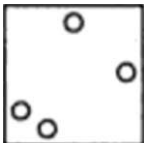
Copyrighted material



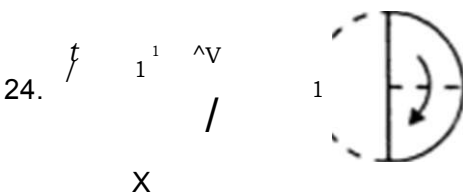
A



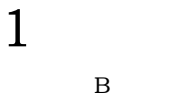
B



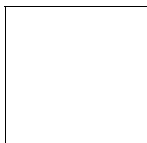
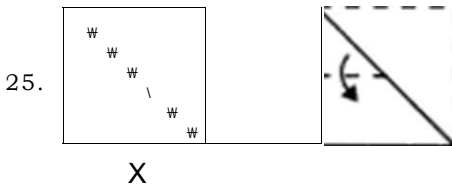
(S.S.C. 1992)



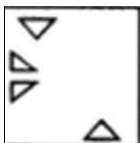
A



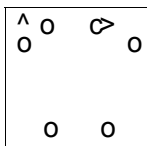
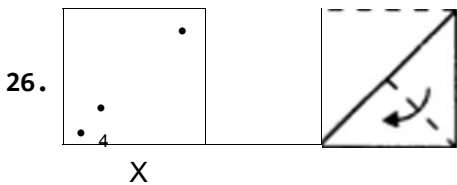
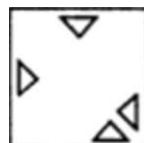
B



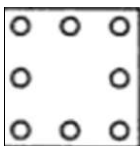
A



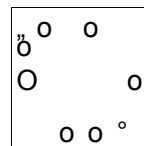
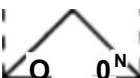
B



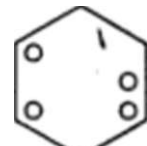
A



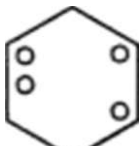
B



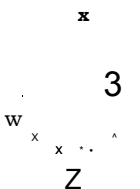
(U.D.C. 1991)



A



B



(I. Tax & Central Excise, 1990)

O_D

28. 4 :

X

U v v J"

=>A A' =

ii v v a

A

Y

U A A If

=•AA' =

JI AA a

B

r "

L -

I -

f -

I

29.

X

Y

30.

X

Y

31.

X

Y

32.

X

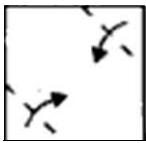
Y

(C.BJ. 1989)

(I. Tax & Central Excise, 1993)

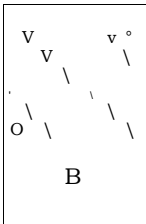
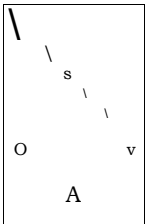
Hidden page

38.

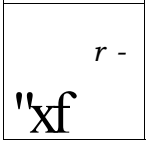


X

Y

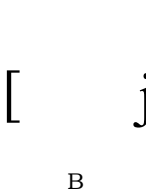
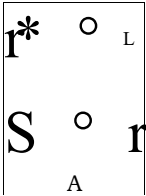


39.



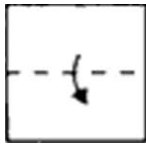
X

Y



(S.S.C. 1995)

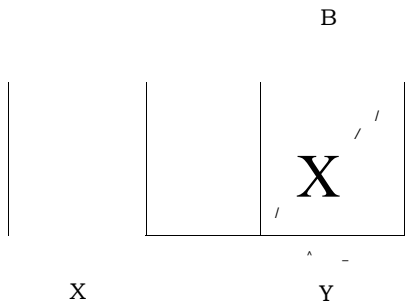
40.



• 3

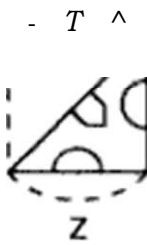
(Asstt. Grade, 1992)

41.



X

Y



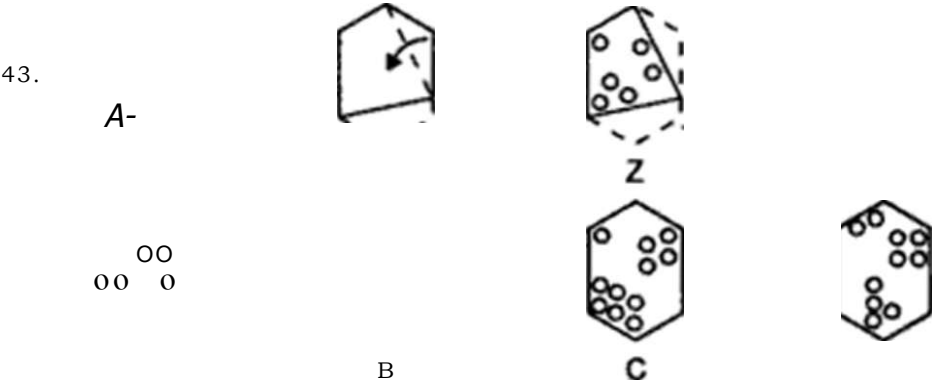
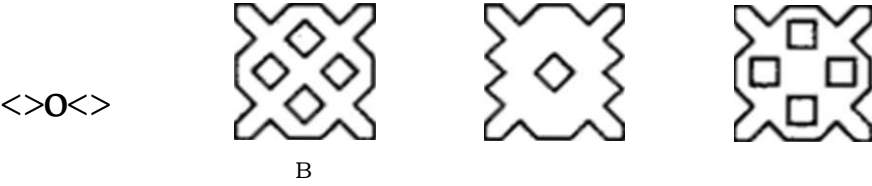
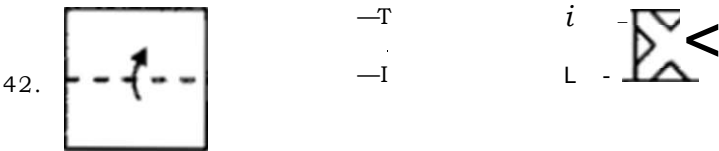
A

B

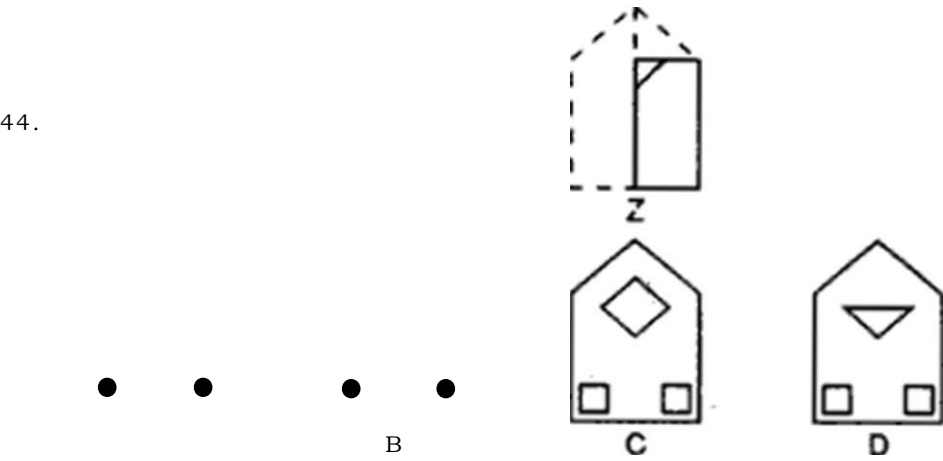


sirr\

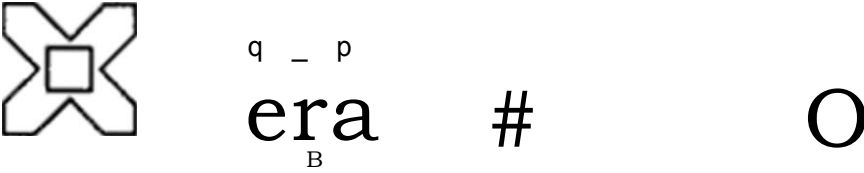
100



(U.D.C. 1995)



(Asstt. Grade, 1996)



46.


X


Y


Z


A


B


C

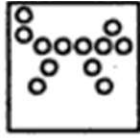

D

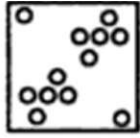
47.

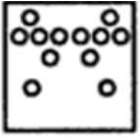

X

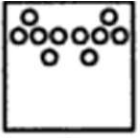

Y


Z


A


B


C


D

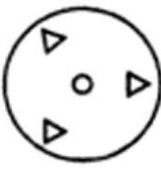
(U.D.C. 1995)

48.


X


Y


Z


A


B


C


D

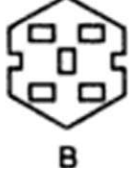
49.

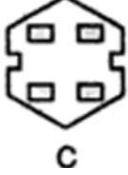

X

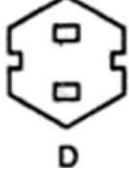

Y


Z


A


B


C


D

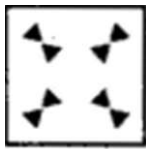
Hidden page

54.

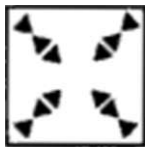


(U.D.C. 1995)

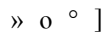
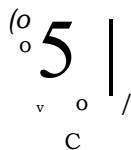
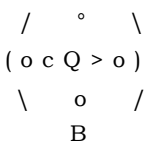
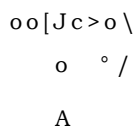
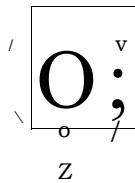
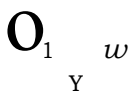
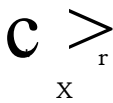
. M



B

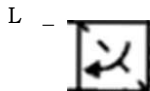
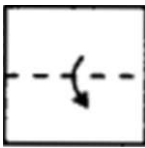


55.



Direction* : In questions 56 to 58, a piece of paper is folded, cut and then unfolded. One of the four alternative figures, marked A, B, C and D, exactly resembles the unfolded paper. Select this figure.

56.



B

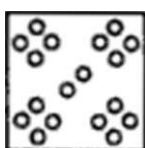
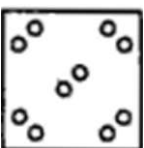
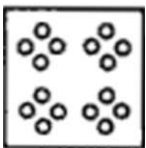
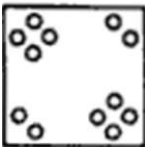


D

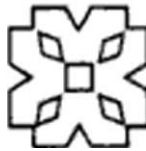
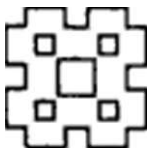
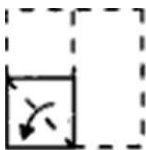
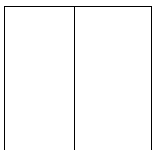
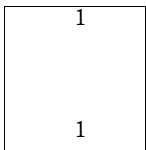
57.



(U.D.C. 1995)



58.



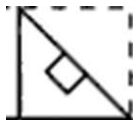
B

C

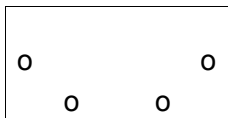
Directions : Each of the questions from 59 to 64 shows a sheet of paper when folded and then cut. This figure is followed by four alternative figures, one of which resembles the sheet when unfolded and has to be selected.

59.

!



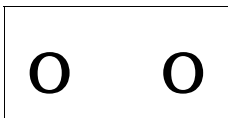
A



B



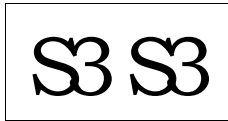
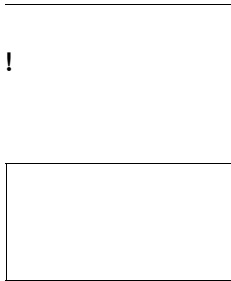
C



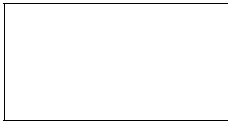
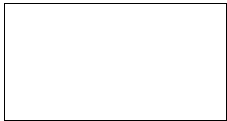
D

60.

!



B

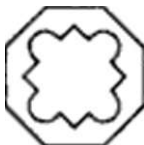


D

61.



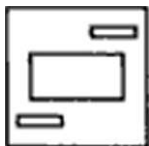
B



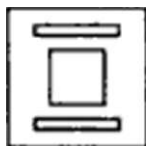
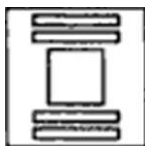
62.



i
t



B



Hidden page